File structure and central contents

This document describes the organization of the repository's files, with the aim of guiding the reader through the location and contents of each folder and file, as well as their relationship to the Jupyter Notebooks.

Main: peace_territories

Inside this folder, you will find the following files and folders:

Readme.md : It provides a brief explanation of the files related to the research project and the article.

License.md: It specifies the license and attribution terms for the data.

DiB_peace_territories.ipynb: A file that includes the descriptive text of the workflow and the executable code for the analysis.

File_structure_and_central_contents.pdf: This is the guide to the location, content, and linking of the files.

Datasets_viewer_and_coding: This folder presents the dataset and the coding used for the analysis, as well as visualization packages for the data. It contains four folders:

databases. It contains the two primary data files (.csv) for analysis in Jupyter Notebooks and a document (.pdf) with the translation

df_vector_hogar_160726.csv This database contains 120,651 household-level records from the "Situational Assessment of Territories of Peace and Equality" questionnaire. The dataset covers 132 variables related to priority groups, household composition, facilities and services, security, and culture, and is supplemented by data on natural hazards from the Risk Atlas and Mexico City's Unified Statistical Registry of Emergency Situations (REUSE).

df_vector_ind_160726_limpia.csv This file contains 153,641 individual records from the "Situational Assessment of Territories of Peace and Equality" questionnaire. The dataset consists of 31 variables covering information on social identity, socioeconomic vulnerability and connectivity, health and well-being, as well as the dynamics of community participation and social cohesion.

Translation of the questions, answers and encoding_TP.pdf. The document contains a breakdown in English of each question and its possible answers in the administered questionnaire, as well as the location of the columns in the files df_vector_ind_160726_limpia and df_vector_hogar_160726.

Geopackage. This folder contains a GeoPackage file (.gpkg) that brings together the vector and raster hazard layers from the Risk Atlas and REUSE, the boundaries of the Territories of Peace polygons, and the heat maps derived from spatial analysis. The file also stores a saved QGIS project with the layer order and styling. To facilitate visual interpretation, it is recommended to add a web basemap once the file has been loaded into the GIS software.

GIS_Peace_territories_cdmx_2026.gpkg:
To load the GeoPackage together with its saved project, run the following code in the QGIS Python console. Note that `clear()` closes the project currently open, so run it in a new, empty QGIS session.

```
from qgis.core import QgsProject

gpkg = r"C:\temp\GIS_Peace_territories_cdmx_2026.gpkg"  #Change the file path
QgsProject.instance().clear()
ok = QgsProject.instance().read(f"geopackage:{gpkg}?projectName=mi_proyecto")
print("Proyecto abierto:", ok)
```

- **23_territories_of_peace_CDMX.** Vector layer containing the polygonal delimitation of the 23 peace territories in Mexico City.
- **62_polygons_territories_of_peace_CDMX.** Vector layer containing the operational subdivision of the 62 fragility pockets comprising the peace territories.
- **62_polygons_territories_of_peace_cdmx_reuse_diag.** Vector layer of the 62 intervention polygons; integrates census diagnosis results and responded attended emergency counts within its attribute table.
- **household_criticality_heatmap.** Density raster model (heatmap) representing the spatial concentration of critical and high-priority households across peace territories.
- **IVT_plus_hazard_risk_index.** Vector layer resulting from the integration of the Territorial Vulnerability Index (IVT) at the polygon scale with the Hazard Risk Index at the Basic Geostatistical Area (AGEB) level.

Emergency records (REUSE, 2018–2025)

- **REUSE_SGIRPC.** Point vector layer providing consolidated coverage of emergency reports from the Ministry of Comprehensive Risk Management and Civil Protection

(SGIRPC), categorized by hazard type (geological, hydrometeorological, chemical-technological, sanitary-ecological, and socio-organizational).

- **geological_phenomena_REUSE.** Point vector layer of emergency records associated with geological phenomena (sinkholes, slope instability, earthquakes, and subsidence).
- **hydrometeorological_phenomena_REUSE.** Point vector layer of emergency records associated with hydrometeorological phenomena (torrential rain, flooding, high winds, hail, and sleet).
- **socio_organizational_phenomena_REUSE.** Point vector layer of emergency records associated with socio-organizational events (accidents, mass gatherings, utility service disruptions, and vandalism).
- **gas_incidents_REUSE.** Point vector layer focused specifically on gas leak emergency reports.
- **electrical_incidents_REUSE.** Point vector layer focused specifically on power grid failures and electrical emergencies.
- **mc_geol_phen_REUSE.** Raster heatmap displaying the spatial density of geological emergencies in CDMX.
- **mc_hydromet_phen_REUSE.** Raster heatmap displaying the spatial density of hydrometeorological emergencies in CDMX.
- **mc_socio_org_phen_REUSE.** Raster heatmap displaying the spatial density of socio-organizational emergencies in CDMX.

Risk Atlas layers (AGEB level)

- **hazard_Risk_index_ATLAS, precipitation_hazard_ATLAS, flood_hazard_ATLAS, earthquake_hazard_ATLAS, slope_susceptibility_hazard_ATLAS.** Vector layers from the Risk Atlas integrating hazard characterizations by category (precipitation, flood, earthquake, and slope susceptibility) along with the Weighted Hazard Index.
- **frequency_of_flooding_2019_2024_ATLAS.** Vector layer from the Risk Atlas systematizing the historical recurrence of street-level waterlogging and flooding (2019–2024).

 figures. Contains the graphical representations and maps derived from the processing and analysis of the data.

 coding_peace_territories_analysis. This folder contains a set of Python scripts for data processing, calculating descriptive metrics, and machine learning-based modeling for territorial diagnosis.

 correlation.py

 critical_sample_size.py

 decision_surface.py

 dispersion.py

 parsing.py

 pca.py

 random-forest.py

 risk.distribution.py

 risk_vulnerability_index.py

 svc-short.py

 time_frame.py